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REVIEW

Which health benefits of the Mediterranean diet are actually supported?

REFERENCES	TOKENS	VERIFICATION	COMPILED
97 verified	≈20.1k	Crossref	26 Aug 2026

ABSTRACT

Mediterranean dietary programmes have their strongest clinical evidence in cardiovascular prevention. In adults at increased risk, three endpoint trials and modern evidence syntheses indicate fewer major cardiovascular events, although absolute benefits are modest and Mediterranean programmes are not clearly superior to every well-supported low-fat programme. Type 2 diabetes is the next most coherent outcome: randomized prevention data, prospective cohorts, and objective biomarker studies point in the same direction, while established diabetes shows small improvements in glycaemia and body mass. Higher adherence is consistently associated with longer survival, but no mortality trial can establish that the observational estimate is causal. Evidence for cancer and dementia is dominated by cohorts; the largest dedicated cognitive trial was null. Small depression trials are favourable but heterogeneous, and fatty-liver benefits appear comparable to those of low-fat diets unless calorie restriction, activity, or polyphenol enrichment is added. The pattern changes inflammation, lipids, metabolites, and gut microbial functions, yet these intermediate responses cannot substitute for clinical outcomes. Overall, the Mediterranean diet is best understood as a supported cardiometabolic programme whose effectiveness depends on adherence and implementation, not as a collection of therapeutic foods or a universal preventive treatment.

01 Introduction

The Mediterranean diet occupies an unusual position in nutrition research because several of its major claims have been tested against clinical endpoints rather than only weight, cholesterol, or self-reported disease. Its evidence base includes randomized controlled trials of primary and secondary cardiovascular prevention, long prospective follow-up in multiple countries, and mechanistic trials that monitor adherence through biomarkers and food provision. At the same time, the label covers variable interventions, and much of the literature on mortality, cancer, dementia, and mental health remains observational. The central question is therefore not whether the pattern is “healthy” in the abstract, but which outcomes it changes, by how much, and under which comparison conditions. Karam et al. 2023, Nucci et al. 2026, English et al. 2021

The distinction between a diet and a supported programme is consequential. Endpoint trials supplied foods, dietitian contact, group sessions, and repeated monitoring, while newer studies add energy restriction, physical activity, or behavioural support. Estruch et al. 2018, Delgado-Lista et al. 2022, Ruiz-Canela et al. 2025 Reviews of dietary patterns also find far more cohort studies than randomized mortality evidence, making healthy-user confounding a persistent alternative explanation for many broad associations. English et al. 2021 A rigorous interpretation must give randomized events more weight than biological plausibility, and prospective associations more weight than cross-sectional correlations, without treating any one trial as infallible.

02 Definition and measurement

The intervention is a family of related eating patterns rather than a prescribed menu. Reviews consistently identify a plant-rich core—vegetables, fruit, legumes, nuts, minimally refined cereals—combined with olive oil as the main added fat, regular fish, and lower consumption of red or processed meat and sweets. Dairy, eggs, alcohol, portion sizes, and frequencies vary across historical and modern definitions. Davis et al. 2015, Bach-Faig et al. 2011 The Seven Countries Study helped formulate the hypothesis through long-term contrasts among populations, but its ecological and cohort observations were not randomized evidence. Menotti & Puddu 2015

Most studies convert the pattern into an adherence score. Some scores award points relative to sex-specific or study-specific medians, while others use fixed intake targets; trials often use the 14-item Mediterranean Diet Adherence Screener (MEDAS). Bach et al. 2006, Schröder et al. 2011 These tools make a whole-diet exposure tractable, but they are not interchangeable: a two-point difference can represent different food substitutions in different datasets, and scoring systems developed in Mediterranean populations may classify non-Mediterranean diets differently. Hutchins-Wiese et al. 2022 Direct comparison within the MCC-Spain study found limited agreement among adherence indexes, showing that instrument choice can change who is labelled highly adherent. Olmedo-Requena et al. 2019

Synthesis quality compounds this exposure problem. An appraisal of 24 cardiovascular reviews found that complete satisfaction of adapted methodological criteria ranged from 8% to 75%; many reports omitted full search details, participant characteristics, or appropriate pooling methods. Huedo-Medina et al. 2016 The implication is not that meta-analysis is uninformative, but that the label “systematic review” does not by itself resolve variation in diet definition, comparator, intervention support, or outcome measurement.

The stable core, variable definitions, and unsupported shortcuts are separated in Figure 1.

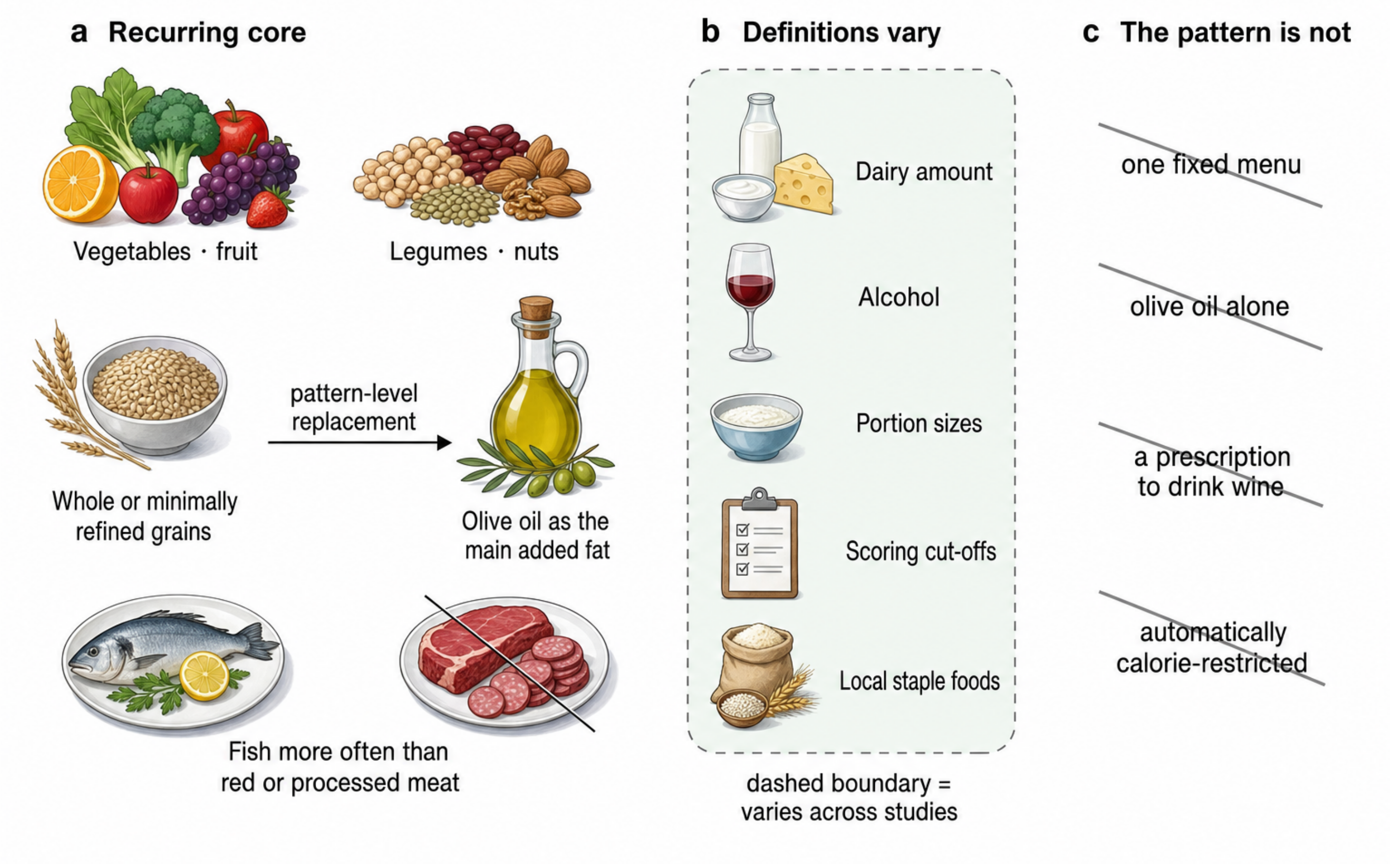

Figure 1. A Mediterranean diet is a pattern, not one fixed menu. The recurring food-group pattern is separated from definition-dependent choices and unsupported single-ingredient or fixed-menu interpretations. The replacement arrow is pattern-level synthesis, not a serving prescription or evidence that one component causes the outcomes. Davis et al. 2015, Bach-Faig et al. 2011, Hutchins-Wiese et al. 2022, Keshani et al. 2024

03 Cardiovascular endpoint trials

PREDIMED provides the largest randomized primary-prevention dataset, but only the corrected republication can be treated as evidence. A statistical investigation of 5,087 randomized trials detected improbable baseline distributions across journals, contributing to scrutiny of prominent reports. Carlisle 2017 PREDIMED investigators subsequently identified household members enrolled without individual randomization and two sites where allocation procedures departed from protocol. The 2013 report was retracted; the 2018 republication analysed the data without assuming fully individual randomization and added exclusions and cluster-robust sensitivity analyses. Estruch et al. 2018

In the corrected analysis, 7,447 adults at high cardiovascular risk were followed for a median of 4.8 years. Compared with low-fat advice, the primary composite of myocardial infarction, stroke, or cardiovascular death had a hazard ratio (HR) of 0.69 (95% confidence interval (CI) 0.53–0.91) for the extra-virgin-olive-oil arm and 0.72 (0.54–0.95) for the nut arm. Excluding 1,588 participants with known or suspected allocation departures yielded essentially the same estimates. Estruch et al. 2018 The correction preserves a clinically important result while reducing the certainty that would attach to an entirely conventional individual randomized controlled trial.

Two secondary-prevention trials extend the finding beyond one study. The Lyon Diet Heart Study randomized 605 people after myocardial infarction and reported cardiac death or nonfatal infarction in 14 Mediterranean-pattern participants and 44 controls over 46 months. de Lorgeril et al. 1999 Its effect was strikingly large and has not been reproduced at the same magnitude. CORDIOPREV randomized 1,002 patients with established coronary disease to Mediterranean or low-fat diets; after seven years, 87 versus 111 participants experienced a major event, corresponding to adjusted hazard ratios around 0.72–0.75. Delgado-Lista et al. 2022 The three settings therefore agree on direction, while trial size, intervention composition, and effect magnitude differ.

Trial	Setting	Comparison	Main endpoint result
PREDIMED republication	High-risk primary prevention; n=7,447	Olive oil or nuts vs low-fat advice	HR 0.69 and 0.72 for major cardiovascular events Estruch et al. 2018
Lyon Diet Heart	Post-infarction; n=605	Mediterranean pattern vs prudent Western diet	Cardiac death/nonfatal infarction: 14 vs 44 de Lorgeril et al. 1999
CORDIOPREV	Coronary disease; n=1,002	Mediterranean vs low-fat diet	87 vs 111 major events; adjusted HR about 0.72–0.75 Delgado-Lista et al. 2022

The exact trial values and the limits on comparing them are shown in Figure 2.

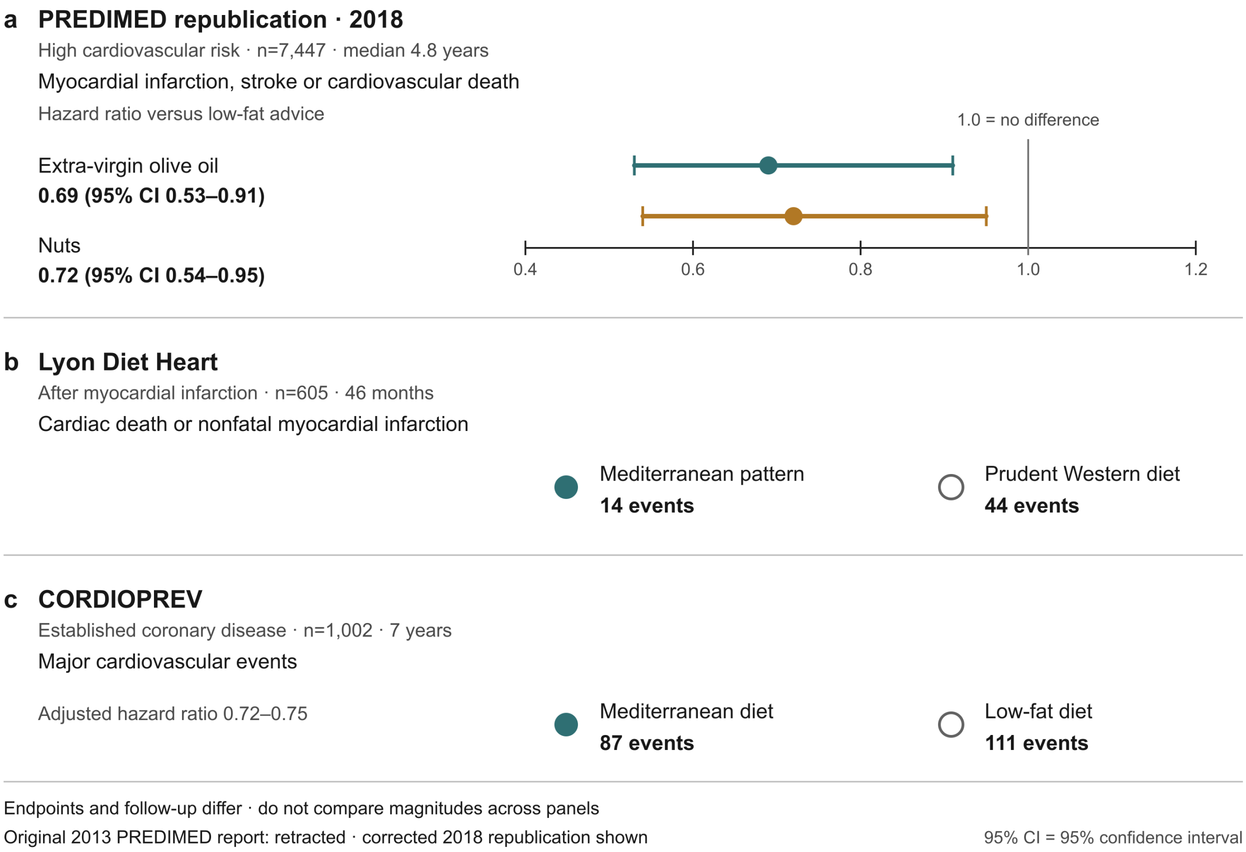

Figure 2. Three endpoint trials support cardiovascular benefit. The proportional panel shows corrected PREDIMED hazard ratios and confidence intervals, while equal-size markers accompany the reported Lyon Diet Heart and CORDIOPREV event counts. Populations, comparators, endpoints, and follow-up differ, so cross-panel magnitude comparisons are invalid; the original 2013 PREDIMED report is retracted, and only the corrected 2018 republication is plotted. Estruch et al. 2018, de Lorgeril et al. 1999, Delgado-Lista et al. 2022, Karam et al. 2023

04 Comparative cardiovascular evidence

The most informative modern synthesis compares supported programmes with both minimal advice and other active diets. A network meta-analysis of 40 trials and 35,548 higher-risk participants estimated that Mediterranean programmes reduced all-cause mortality (odds ratio 0.72), cardiovascular mortality (0.55), stroke (0.65), and nonfatal infarction (0.48) versus minimal intervention. At intermediate baseline risk, the estimates corresponded to about 17 fewer deaths, 13 fewer cardiovascular deaths, seven fewer strokes, and 17 fewer infarctions per 1,000 people over five years. Karam et al. 2023 However, low-fat programmes also reduced mortality and infarction, and there was no convincing advantage of Mediterranean over low-fat programmes for those endpoints. The result supports structured dietary care more strongly than dietary exceptionalism.

Earlier syntheses were more cautious. Six randomized trials with 10,950 participants produced a risk ratio of 0.63 for major vascular events but 1.00 for all-cause mortality; excluding one trial with serious integrity concerns erased several favourable secondary findings. Liyanage et al. 2016 A Cochrane review included 30 trials and 12,461 participants and emphasized low-to-moderate certainty, heterogeneous definitions, sparse events, and weak adverse-event reporting. Rees et al. 2019 Another systematic review concluded that unrestricted-fat Mediterranean patterns improved several health outcomes, while noting that the clinical-event evidence depended heavily on a small number of trials. Bloomfield et al. 2016

Newer reviews continue to find fewer cardiovascular events without removing the structural dependence on PREDIMED. Sebastian et al. 2024 The 2026 Italian-guideline syntheses included 87 primary-prevention studies with more than 1.4 million participants and 19 secondary-prevention studies with more than 91,000; they reported lower incident and recurrent cardiovascular risk, but combined randomized and cohort estimates and graded certainty separately by outcome. Volpe et al. 2026a, Volpe et al. 2026b Risk-factor comparisons are smaller than event comparisons: Mediterranean diets modestly improve weight, pressure, glucose, and lipids relative to low-fat diets, while active-diet differences are narrower than contrasts with usual care. Nordmann et al. 2011

05 Mortality and longevity

Prospective cohorts show a remarkably stable dose-response association between Mediterranean adherence and survival. In 22,043 Greek adults, each two-point increase in the original score corresponded to a 25% lower mortality rate; in 74,607 older participants across Europe, the analogous estimate was 8%. Trichopoulou et al. 2003, Trichopoulou et al. 2005 The difference between those estimates is informative: the association survives geographic expansion but becomes more modest.

Early pooling of 12 cohorts and 1.57 million participants estimated all-cause mortality risk ratio 0.91 per two score points. Sofi et al. 2008 Subsequent syntheses remained close to a 9%–10% reduction per two points despite differences in score and follow-up. Eleftheriou et al. 2018, Soltani et al. 2019 The most recent broad meta-analysis screened 5,229 records and included 54 cohorts, 1.83 million participants, and 346,034 deaths. Its pooled risk ratio was 0.96 (95% CI 0.95–0.97) per one-point adherence increase, with moderate certainty. Nucci et al. 2026 This is a graded association, not evidence that a single menu reduces mortality by a fixed percentage.

Large US cohorts support transfer beyond the Mediterranean. In the Nurses' Health Study and Health Professionals Follow-up Study, the top healthy-pattern categories were associated with total mortality HR 0.82 and cardiovascular disease HR 0.83. Shan et al. 2023, Shan et al. 2020 Among 25,315 US women followed for up to 25 years, top adherence corresponded to mortality HR 0.77; metabolite, inflammatory, triglyceride-rich lipoprotein, insulin-resistance, and body-mass pathways explained only part of the estimate. Ahmad et al. 2024 People with diabetes also show an inverse association, with mortality HR 0.63 per two score points in the Moli-sani cohort. Bonaccio et al. 2016

Objective exposure measures help but do not convert cohorts into trials. In 642 older Italians with 20-year follow-up, a biomarker-based score predicted all-cause mortality HR 0.72 and cardiovascular mortality HR 0.55 for the highest versus lowest tertile, whereas repeated food-questionnaire scoring was weaker. Agreement between biomarker and questionnaire scores was low, biomarkers were measured only at baseline, and residual confounding remained possible. Hidalgo-Liberona et al. 2021 The mortality literature is therefore consistent and dose-responsive, but its causal magnitude remains uncertain because no trial was powered to reproduce the cohort endpoint.

06 Diabetes, weight, and metabolic health

Type 2 diabetes is the non-cardiovascular outcome for which experimental, prospective, and biomarker evidence align most closely. In the diabetes-free PREDIMED subgroup, both Mediterranean interventions lowered incidence relative to low-fat advice. Salas-Salvadó et al. 2014 An analysis using up to eight annual adherence measurements found HR 0.80 (95% CI 0.70–0.92) per two MEDAS points and 0.46 (0.25–0.83) for the highest versus lowest adherence category. Martínez-González et al. 2023 Because attained adherence is partly behavioural rather than purely randomized, these estimates are strongest as evidence of a gradient within a trial context.

PREDIMED-Plus tested whether an intensive weight-loss programme added benefit. Among 4,746 adults free of diabetes, energy-restricted Mediterranean diet plus activity reduced six-year incidence by 31% relative to an unrestricted Mediterranean diet: 9.5% versus 12.0%. Ruiz-Canela et al. 2025 This comparison is clinically useful but cannot isolate diet, energy deficit, exercise, and behavioural contact. In a separate 1,521-person imaging substudy, the same programme reduced total and visceral fat and attenuated relative lean-mass decline over three years, with partial weight regain after the first year. Konieczna et al. 2023

Objective biomarkers reinforce the diabetes association. A score based on five carotenoids and 24 fatty acids discriminated the Mediterranean arm of a controlled feeding trial with a C-statistic of 0.88, then predicted incident diabetes HR 0.71 per standard deviation in 22,202 EPIC-InterAct participants, versus HR 0.90 for self-report. Sobiecki et al. 2023 The score may capture foods outside the Mediterranean pattern and still permits residual confounding, but it makes reporting bias alone an unlikely explanation. A broader Mediterranean-lifestyle index also predicted diabetes HR 0.70 for the highest versus lowest quartile in 112,493 UK Biobank adults, although the exposure included activity, rest, and social habits. Maroto-Rodriguez et al. 2023

Treatment effects in established diabetes are smaller. Prospective and trial reviews estimate lower incidence with adherence and modest improvements in glycaemic control relative to comparator diets. Schwingshackl et al. 2015, Esposito et al. 2015 A 2025 meta-analysis of 11 trials found glycated haemoglobin (HbA1c) lower by 0.31 percentage points, fasting glucose by 0.85 mmol/L, and body mass index by 0.83 kg/m^2. Wu et al. 2025 These changes are useful adjuncts, not evidence that the diet replaces medication.

Weight-loss comparisons likewise argue against uniqueness. DIRECT randomized 322 adults and found two-year losses of 4.4 kg with Mediterranean, 2.9 kg with low-fat, and 4.7 kg with low-carbohydrate diets. Shai et al. 2008 Across 121 randomized trials, most named diets produced modest six-month loss that diminished by 12 months; Mediterranean programmes were notable for retaining some cardiovascular risk-factor improvement, not for superior long-term weight loss. Ge et al. 2020 A 50-study synthesis and a 57-trial review found small improvements across metabolic-syndrome components, with considerable heterogeneity. Kastorini et al. 2011, Papadaki et al. 2020 A green Mediterranean variant more than doubled visceral-fat loss relative to standard Mediterranean diet, but added walnuts, green tea, Mankai, energy restriction, and activity in a workforce that was 88% male. Zelicha et al. 2022

The null evidence identifies the appropriate unit of intervention. A 2024 meta-analysis of olive-oil-enriched Mediterranean trials found no consistent cardiometabolic advantage except a very small triglyceride reduction, so adding oil does not reproduce the whole pattern. Keshani et al. 2024 In people with pre-existing metabolic disease, a 69-study synthesis found modest improvement in body mass index, waist, low-density lipoprotein (LDL) cholesterol, triglycerides, fasting glucose, insulin resistance, and C-reactive protein (CRP), while HbA1c and lean-mass results were inconsistent. Muscogiuri et al. 2026

07 Cancer

Cancer is where breadth of association most clearly exceeds causal depth. An umbrella review judged evidence comparatively robust for overall cancer incidence and mortality but only suggestive or weak for many individual sites. Dinu et al. 2018 The largest updated synthesis included 117 studies and 3.20 million participants. High adherence was associated with cancer mortality risk ratio 0.87, colorectal cancer 0.83, breast cancer 0.94, and larger reductions for some less common sites, yet most site-specific comparisons were rated low or very low certainty. Morze et al. 2021

Repeated meta-analyses find the same broad direction but share the same observational inputs, scoring heterogeneity, and residual lifestyle confounding. Schwingshackl et al. 2017 Dose-response analysis supports lower colorectal risk, and a cohort/meta-analysis of postmenopausal breast cancer found the inverse association concentrated in estrogen-receptor-negative tumours. Zhong et al. 2020, van den Brandt & Schulpen 2017 These subgroup patterns can be biologically interesting without being intervention effects.

Randomized evidence consists mainly of one small secondary outcome. In PREDIMED, the olive-oil group experienced fewer invasive breast cancers than low-fat controls, HR 0.32 (95% CI 0.13–0.79), but only 35 cases occurred and cancer was not the trial's primary endpoint. Toledo et al. 2015 A breast-cancer umbrella review reaches a favourable overall interpretation while acknowledging the predominance of observational evidence. González-Palacios Torres et al. 2023 Olive-oil intake itself is associated with lower cardiovascular disease, diabetes, and mortality, but its pooled cancer estimate is null at risk ratio 0.94 (0.86–1.03). Martínez-González et al. 2022 The cancer literature therefore supports prevention hypotheses and a generally plant-rich pattern, not a claim that Mediterranean diet treatment prevents specific cancers.

08 Cognition and dementia

Prospective studies associate Mediterranean adherence with lower cognitive decline. A 2025 meta-analysis of 23 studies reported HR 0.82 for cognitive impairment, 0.89 for dementia, and 0.70 for Alzheimer disease, while documenting heterogeneity, possible publication bias, and limited non-Western representation. Fekete et al. 2025 A separate updated synthesis also found inverse dementia and Alzheimer associations. Nucci et al. 2024 UK Biobank provides a useful absolute-risk estimate: among 60,298 adults, high adherence corresponded to dementia HR 0.77 and absolute risk 1.18% versus 1.73% over 9.1 years, without measurable interaction with polygenic risk. Shannon et al. 2023

Earlier reviews combined favourable cohorts with small trials. One synthesis found mild cognitive impairment risk ratio 0.75 and Alzheimer disease 0.71 in cohorts; two trials yielded an episodic-memory standardized mean difference of 0.20. Fu et al. 2022 Another meta-analysis reached a similar observational conclusion. Singh et al. 2014 A PREDIMED-Navarra substudy reported better global cognition after 6.5 years, but it was small and inherited the allocation limitations of the parent trial. Martínez-Lapiscina et al. 2013

The largest dedicated trial sharply lowers expectations. The three-year MIND trial randomized 604 older adults with suboptimal diets and obesity history; both intervention and control were calorie-restricted. Global cognition differed by only 0.035 standard deviations (95% CI −0.022 to 0.092), and brain magnetic resonance imaging (MRI) outcomes did not differ. Barnes et al. 2023 A MIND-diet meta-analysis remains favourable because its pooled estimate is dominated by cohorts, whereas a 2024 PREVENT analysis found no association with cognition or neuroimaging. Chen et al. 2023, Gregory et al. 2024 The most defensible synthesis is that long-term adherence may mark lower dementia risk, but a three-year supported MIND intervention did not improve cognition in relatively healthy older volunteers.

09 Depression and mental health

Prospective cohorts associate Mediterranean adherence with lower incident depression. Meta-analyses report an inverse association of roughly one-third for the highest adherence groups, although reverse causation is difficult to exclude because early depression can alter shopping, appetite, activity, and diet quality. Lassale et al. 2019, Shafiei et al. 2019 The observational result is therefore compatible with benefit without proving treatment efficacy.

Small randomized trials provide a causal signal but an unstable magnitude. SMILES randomized 67 adults with major depression to dietary support or befriending-style social support for 12 weeks. The diet group improved 7.1 points more on the Montgomery-Åsberg Depression Rating Scale, and remission was 32% versus 8%. Jacka et al. 2017 HELFIMED enrolled 152 adults and found greater symptom improvement with group Mediterranean education plus fish-oil supplementation than with social groups, a design that cannot isolate the diet. Parletta et al. 2019 AMMEND extended the result to 72 young men over 12 weeks, again in a small, selected population. Bayes et al. 2022

Pooling magnifies the uncertainty. Five trials with 1,507 participants yielded symptom standardized mean difference −0.53, but heterogeneity was 87%, the prediction interval crossed zero, and certainty was low. Bizzozero-Peroni et al. 2025 A 2025 review identified 104 mental-health studies—88 observational and only 16 interventional—and found favourable associations with depression, anxiety, stress, quality of life, and well-being. Kabthymer et al. 2026 Direction is consistent enough to justify larger trials, but current evidence does not identify a stable antidepressant effect or support replacing established care.

10 Fatty liver

Mediterranean interventions reduce hepatic fat, but direct comparison weakens claims of unique superiority. In a 12-week randomized trial, ad libitum Mediterranean and low-fat diets both reduced steatosis without a decisive between-diet difference. Properzi et al. 2018 In DIRECT, decreased hepatic fat statistically mediated part of the Mediterranean diet's cardiometabolic advantage over low fat, but mediation cannot identify the responsible foods. Gepner et al. 2019

The largest imaging effects come from an enriched, energy-restricted variant. DIRECT-PLUS randomized 294 adults to healthy guidance, standard Mediterranean, or green Mediterranean diets, with physical activity in every arm. Relative liver-fat reduction was 12.2%, 19.6%, and 38.9%, respectively, after 18 months; the green arm added walnuts, green tea, Mankai, and further red-meat restriction. Yaskolka Meir et al. 2021 This shows that a complex enriched programme can change liver fat, not that olive oil alone treats liver disease.

Meta-analyses explain the mixed interpretation. Earlier randomized-trial pooling found improvement in steatosis and insulin resistance and small, inconsistent enzyme changes. Kawaguchi et al. 2021, Sangouni et al. 2022 A 2024 head-to-head review concluded that Mediterranean and low-fat diets had similar short-term effects on liver enzymes and fat content. Xiong et al. 2024 An Asian-adapted randomized trial suggests the approach can be transferred culturally, but adaptation changes the tested foods and long-term clinical liver outcomes remain unavailable. Chooi et al. 2024

11 Biological pathways

Mechanistic evidence supports plausibility through several pathways rather than one "active ingredient." Trial syntheses report small changes in endothelial function and selected inflammatory markers, with high heterogeneity across assays and populations. Schwingshackl & Hoffmann 2014, Koelman et al. 2022 A 2026 review of 33 randomized trials and 3,476 participants found reductions in high-sensitivity CRP, interleukin-6, and interleukin-17, but not conventional CRP, interleukin-10, tumour-necrosis factor, or total antioxidant capacity. Keshani et al. 2026 Reviews therefore implicate fat quality, plant polyphenols, fibre fermentation, inflammation, endothelial function, and insulin sensitivity without establishing one required component. Itsiopoulos et al. 2022

Controlled feeding and isocaloric designs separate dietary composition from weight loss. In an eight-week trial of 82 overweight or obese adults, a weight-maintaining Mediterranean diet lowered total cholesterol and altered faecal bile acids, microbial functions, and metabolites. Glucose, insulin, and CRP did not change. Meslier et al. 2020 In older adults, the one-year NU-AGE intervention shifted microbial taxa and functions that covaried with lower frailty and inflammation; the analysed microbiome subset included 612 participants across five countries. Ghosh et al. 2020 A separate obesity trial found that both Mediterranean and low-fat healthy diets modified gut communities alongside improved insulin sensitivity. Haro et al. 2016

Molecular profiles are also becoming exposure measures. Plasma metabolomic patterns tracked Mediterranean adherence and subsequent cardiovascular risk, though they cannot by themselves prove mediation. Li et al. 2020 In DIRECT-PLUS, baseline bile acids and particular microbial species modified lipid and adiposity responses, while a PREDIMED-Plus analysis connected intervention-related microbiome and metabolome shifts with risk-factor changes. Gao et al. 2024, García-Gavilán et al. 2024 These studies clarify how people may respond differently; they do not upgrade surrogate changes into evidence for fewer cancers, dementia cases, or deaths.

12 Implementation, cost, and sustainability

Transfer outside Mediterranean settings requires adaptation rather than literal replication. Reviews emphasize matching the pattern to local staples, availability, culinary practice, and cultural identity while preserving its plant-forward structure and fat quality. Hoffman & Gerber 2013, Woodside et al. 2022 A Mediterranean score developed around olive oil, wine, and regional medians may misclassify an equally plant-rich pattern elsewhere. Implementation is therefore a question of function—what foods replace refined grains, processed meat, and saturated fat—not geographic authenticity.

Supported efficacy also exceeds unsupported adoption. MedEx-UK showed that a theory-informed programme could improve adherence and was acceptable to older adults at dementia risk, but it used behavioural maintenance and physical-activity components. Jennings et al. 2024 Cross-country studies identify cost, access, cooking skills, family norms, and nutritional knowledge as barriers. Biggi et al. 2024 PREDIMED supplied oil or nuts; PREDIMED-Plus added frequent dietitian contact and activity coaching. Estruch et al. 2018, Ruiz-Canela et al. 2025 Real-world benefit will decline if those adherence supports disappear.

Economic and environmental effects depend on assumptions. A secondary-prevention analysis estimated Mediterranean diet plus activity to be cost-effective, but the model inherits adherence and event-reduction estimates from other data. Bonekamp et al. 2024 A systematic review of 35 sustainability studies—25 modelling, seven cross-sectional, and three longitudinal—reported the pattern environmentally favourable in 91% of comparisons. Most gains came from lower ruminant-meat intake, while water demand and local production could change particular rankings. Lorca-Camara et al. 2024 The Med Diet 4.0 framework similarly treats health, environment, culture, and local economy as linked dimensions rather than interchangeable outcomes. Dernini et al. 2017

Alcohol requires a modern qualification. Traditional scores often award a point for moderate wine, but alcohol raises cancer, dependence, and injury risk. A contemporary review argues that wine is not necessary to preserve the pattern's benefits and should not be recommended to non-drinkers. Martínez-González 2024 The therapeutic interpretation is therefore vegetables, legumes, whole grains, nuts, fish, and unsaturated fats—not an instruction to begin drinking.

The intervention package, adherence conditions, and the causal stopping point are traced in Figure 3.

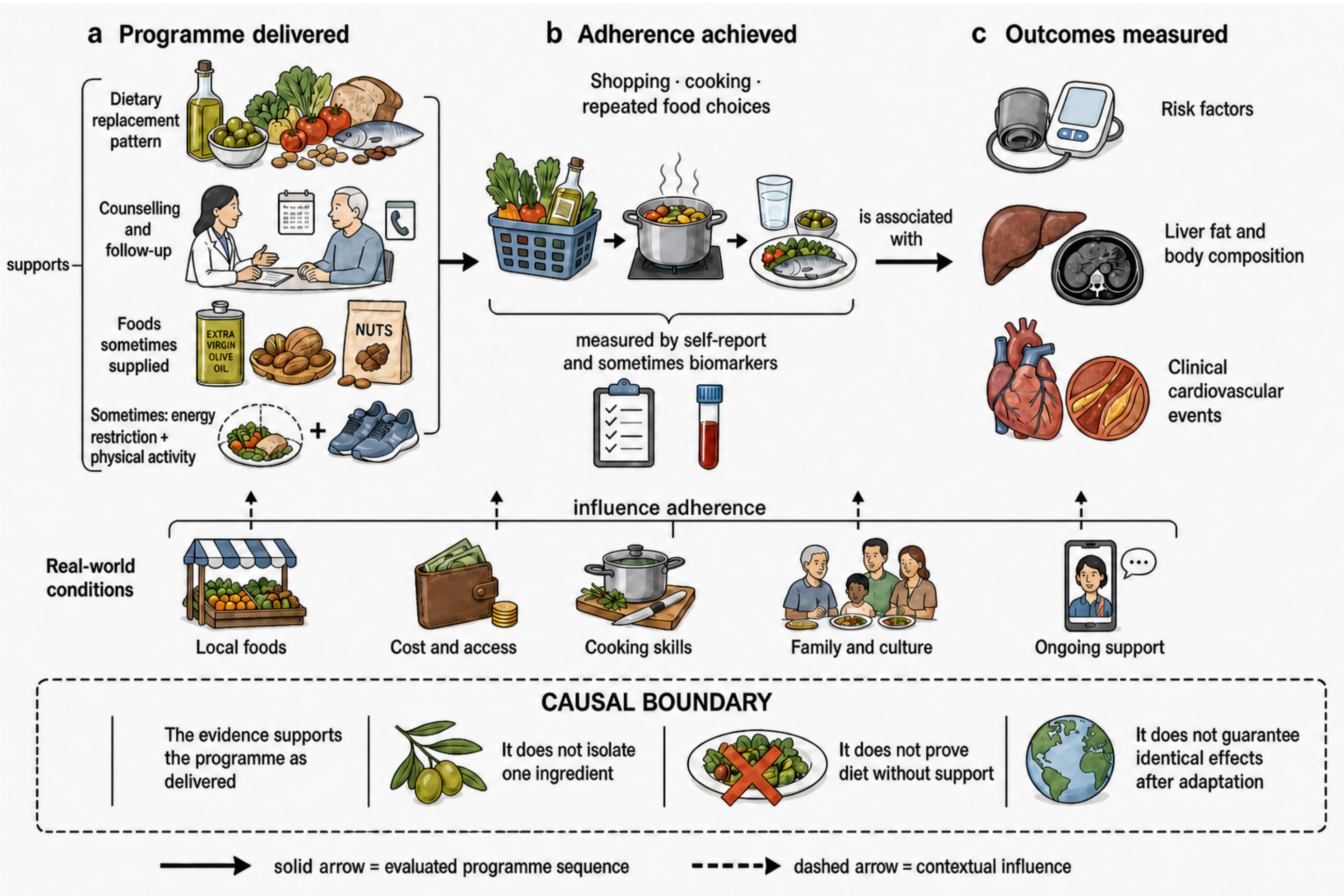

Figure 3. Benefits belong to the supported programme as delivered. Trials evaluate a replacement pattern with varying counselling, food provision, and lifestyle support; real-world conditions influence adherence before risk factors, imaging outcomes, or clinical events are measured. The arrows express programme logic rather than a quantified mediation model, and the dashed boundary prevents attributing the package to one ingredient, unsupported diet alone, or identical effects after cultural adaptation. Estruch et al. 2018, Jennings et al. 2024, Biggi et al. 2024, Lorca-Camara et al. 2024, Ruiz-Canela et al. 2025

13 Overall certainty

Outcome	Current conclusion	Main reason confidence is limited
Major cardiovascular events	Probably reduced in higher-risk adults, particularly versus minimal support	Few endpoint trials; corrected PREDIMED carries much of primary-prevention weight Karam et al. 2023
Type 2 diabetes	Probably prevented; established disease improves modestly	Intensive programmes combine diet, energy restriction, activity, and support Ruiz-Canela et al. 2025

Outcome	Current conclusion	Main reason confidence is limited
All-cause mortality	Consistent 4% lower risk per adherence point in cohorts	No randomized mortality confirmation; residual healthy-user bias Nucci et al. 2026
Weight and risk factors	Small favourable changes	Similar benefits occur with other supported healthy diets Ge et al. 2020
Cancer	Several inverse site-specific associations	Predominantly observational; one small randomized secondary signal Morze et al. 2021
Cognition	Cohorts favourable, largest dedicated trial null	Active control, limited duration, exposure confounding Barnes et al. 2023
Depression	Small trials favourable	Low certainty, high heterogeneity, weak blinding Bizzozero-Peroni et al. 2025
Fatty liver	Liver fat decreases	Short-term equivalence to low-fat diets; few clinical outcomes Xiong et al. 2024

The evidence hierarchy is consequently uneven. Cardiovascular events and diabetes prevention combine randomized and observational support. Mortality has scale, dose-response, and geographic replication but not randomized confirmation. Cancer and dementia are mostly cohort findings, depression is a small-trial signal, and liver studies mainly measure imaging or enzymes. Mechanistic responses make the cardiometabolic findings coherent, but they cannot fill causal gaps for unrelated outcomes.

This uneven directness and certainty are mapped in Figure 4.

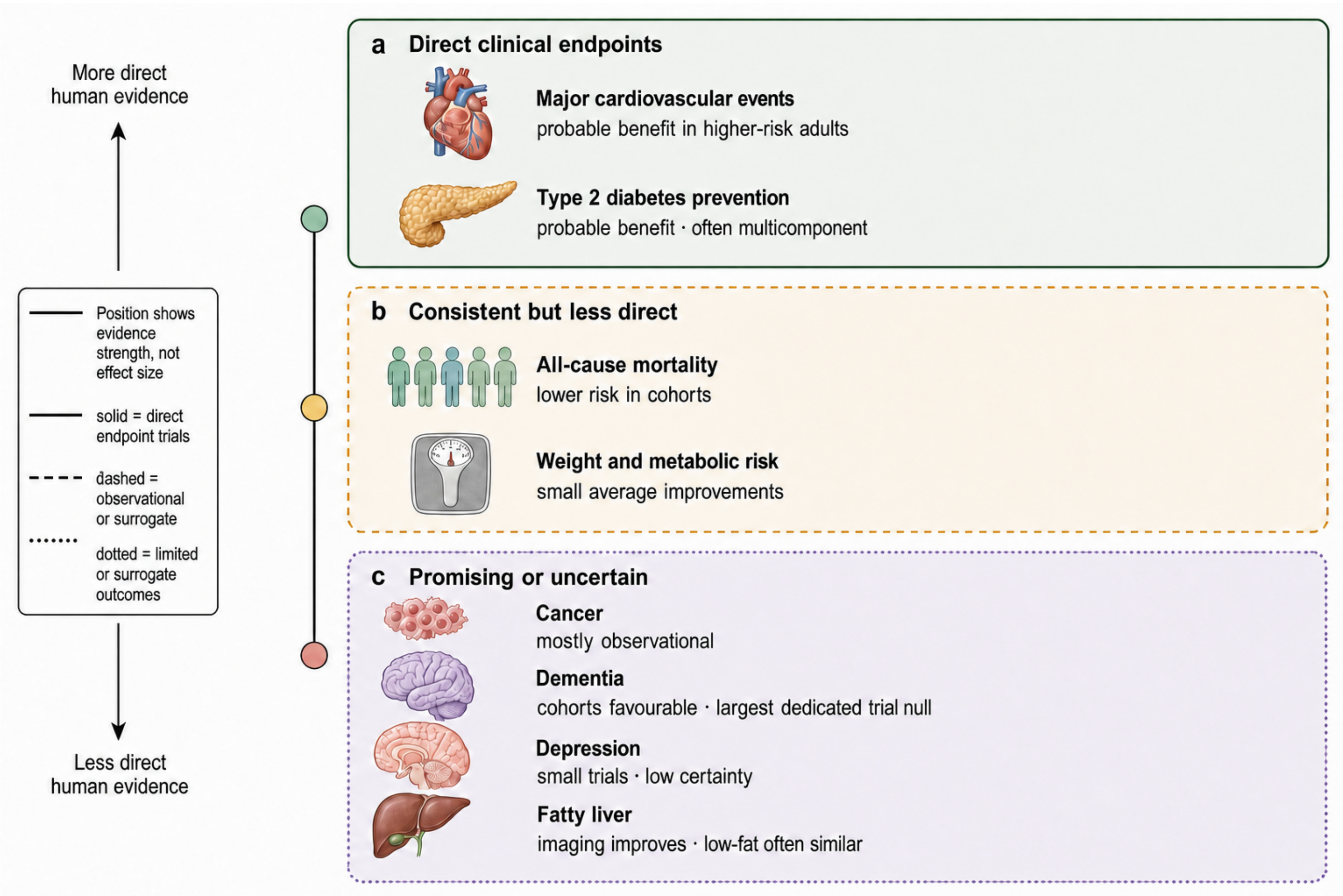

Figure 4. The evidence is strongest for cardiometabolic outcomes. Outcomes are arranged by human evidence directness and certainty, from clinical endpoint trials through consistent but less-direct evidence to limited or surrogate outcomes. This is a narrative-review synthesis rather than a formal GRADE score: position, marker area, and colour do not encode effect size or an individual's expected benefit. Karam et al. 2023, Ruiz-Canela et al. 2025, Nucci et al. 2026, Morze et al. 2021, Barnes et al. 2023

14 Conclusion

The Mediterranean diet has a defensible, bounded set of benefits. Structured programmes probably reduce major cardiovascular events in adults at increased risk, especially compared with minimal advice; they are not clearly superior to every supported low-fat programme. Type 2 diabetes is probably prevented, and established diabetes, body composition, and metabolic risk factors improve by modest amounts. Higher adherence is consistently associated with longer survival, but the size of that effect remains observational.

Claims should be weaker for other outcomes. Cancer associations cover several sites without adequate prevention trials. Dementia cohorts are favourable, yet the largest dedicated cognitive trial was null. Depression trials point toward benefit but are too small and heterogeneous to define a treatment effect. Energy-restricted and enriched variants reduce liver fat, while standard Mediterranean and low-fat diets perform similarly in direct short-term comparison. The practical conclusion is not that olive oil, nuts, wine, or polyphenols are medicines. It is that replacing processed meat, refined foods, and saturated fat with a supported, culturally workable plant-forward pattern has credible cardiometabolic value, with the remaining claims still awaiting stronger trials.

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